

CMPE 49T HW2

Problem 1: Predicting Lesions from Image Patch features

(a) Logistic Model and Likelihood

The logistic model is

$$p(y=1|x) = \sigma(z) = \frac{1}{1+e^{-z}}, \quad z = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b$$

For a single observation (x, y) , the likelihood is

$$\mathcal{L}(w, b | x, y) = p^y (1-p)^{1-y}$$

The negative log-likelihood over N samples gives the cross-entropy loss:

$$\ell(w, b) = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1-y_i) \log(1-p_i)]$$

Gradient derivations:

$$\frac{\partial \ell}{\partial w_j} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i) x_{ij}, \quad \frac{\partial \ell}{\partial b} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i).$$

Example: for $p_i = 0.8$, $y_i = 1$, $x_{ij} = 1$, the contribution to $\partial \ell / \partial w_j$ is $(0.8 - 1) = -0.2$.

(b) Interaction Between Intensity and Contrast

We add an interaction feature $x_5 = x_1 x_2$:

$$P(y=1|X) = \sigma(w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + w_5 x_1 x_2 + b)$$

The decision boundary is given by $p=0.5 \iff$

$z=0$, fixing x_3, x_4 to constants, write

$$w_5 x_1 x_2 + w_1 x_1 + w_2 x_2 + c = 0, \quad c = w_3 x_3 + w_4 x_4 + b$$

This is a conic in (x_1, x_2) of the form

$$A x_1^2 + B x_1 x_2 + C x_2^2 + D x_1 + E x_2 + F = 0 \quad \text{with } A=C=0$$

and $B = w_5$. For $w_5 \neq 0$, the conic is a hyperbola;

if $w_5 = 0$ it degenerates to a line $w_1 x_1 + w_2 x_2 + c = 0$

Solving for x_2 (when $w_5 x_1 + w_2 \neq 0$) makes the hyperbolic nature explicit:

$$x_2(x_1) = -\frac{w_1 x_1 + c}{w_5 x_1 + w_2}, \quad \text{whose asymptotes are}$$

$$x_1 = -\frac{w_2}{w_5} \quad \text{and} \quad x_2 = -\frac{w_1}{w_5}, \quad \text{obtained by examining}$$

the denominator zero and the large $|x_1|$ limit.

The interaction weight w_5 controls how the marginal effect of one feature depends on the other.

$$\frac{\partial z}{\partial x_1} = w_1 + w_5 x_2, \quad \frac{\partial z}{\partial x_2} = w_2 + w_5 x_1$$

If $w_5 > 0$, increasing x_2 amplifies the effect of x_1 , so high intensity together with high contrast raises lesion probability more than additively.

(C) Elastic Net Regularization (Clarified)

We regularize only the weights w (not the bias b).

Let $R(w) = \lambda [\alpha \|w\|_1 + (1-\alpha) \|w\|_2^2]$, with b unregularized.

The total loss is $\ell_{\text{tot}}(w, b) = \ell(w, b) + R(w)$.

With this convention

$$\frac{\partial \ell_{\text{tot}}}{\partial w_j} = \frac{\partial \ell}{\partial w_j} + \lambda [\alpha \text{sign}(w_j) + 2(1-\alpha)w_j],$$

$$\frac{\partial \ell_{\text{tot}}}{\partial b} = \frac{\partial \ell}{\partial b}$$

At $w_j = 0$, the subgradient of $|w_j|$ is the interval $[-1, 1]$.

(d) One gradient Descent Update Example:

We define the interaction $x_5 = x_1 x_2$. Initialize $w_j = 0.1$ for all j , $b = 0$, step size $\eta = 0.5$, and L2 strength $\lambda = 0.2$. Probabilities $p_i = \sigma(z_i)$ below are computed from these initial parameters. We do not penalize b .

Patch	x_1	x_2	x_3	x_4	x_5	z_i	p_i	y_i	$p_i - y_i$
1	1	0	1	1	0	0.3	0.5744	1	-0.4256
2	0	1	1	0	0	0.2	0.5498	0	0.5498
3	1	1	0	1	1	0.4	0.5987	1	-0.4013
4	0	0	1	0	0	0.1	0.5250	0	0.5250
5	1	0	0	1	0	0.2	0.5498	1	-0.4502

Average gradients:

$$\frac{\partial \ell}{\partial w} = \begin{bmatrix} -0.2554 \\ 0.0297 \\ 0.1298 \\ -0.2554 \\ -0.0803 \end{bmatrix}, \quad \frac{\partial \ell}{\partial b} = -0.0405$$

After adding $2\lambda w = 0.04$, total gradients:

$$\nabla_w \ell_{\text{tot}} = \begin{bmatrix} -0.2154 \\ 0.0697 \\ 0.1698 \\ -0.2154 \\ -0.0403 \end{bmatrix}$$

On update with $\eta = 0.5$ gives:

$$w^{(1)} = \begin{bmatrix} 0.2077 \\ 0.0651 \\ 0.0251 \\ 0.2077 \\ 0.1202 \end{bmatrix}, \quad b^{(1)} = 0.0203$$

(d) - Continue

Interpretation: positive weights w_1, w_4, w_5 strengthen their features contribution, while small or negative gradients reduce less informative ones.
Regularization keeps magnitudes modest and stabilizes learning.

$$\frac{\partial \ell}{\partial w} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i) x_i, \quad \frac{\partial \ell}{\partial b} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)$$

With pure L2 ($\alpha = 0$), the total gradient adds $2\lambda w$:

$$\nabla_w \ell_{\text{tot}} = \frac{\partial \ell}{\partial w} + 2\lambda w, \quad \nabla_b \ell_{\text{tot}} = \frac{\partial \ell}{\partial b}$$

We then update

$$w^{(1)} = w^{(0)} - \eta \nabla_w \ell_{\text{tot}}, \quad b^{(1)} = b^{(0)} - \eta \nabla_b \ell_{\text{tot}}$$

(e) Regularization Strength and Model Complexity

- As λ increases, the average $|w_j|$ decreases monotonically toward zero. Training accuracy decreases for high λ (underfitting), while validation accuracy first rises then drops. Best generalization occurs at an intermediate λ . Given $w_1 = 1.2$, $w_2 = 0.9$, $w_3 = 0.02$, $w_4 = 0.01$, $w_5 = 0$, we infer texture variance and edge density have little influence.

(f) Model Interpretation

- final model : $w_1 = 1.35$, $w_2 = -0.95$, $w_3 = 0.02$, $w_4 = 0.10$, $w_5 = 0$, $b = -0.15$. Feature importance $x_1 > x_2 > x_4 > x_3 > x_5$. Positive w_1 means higher intensity increases lesion probability; negative w_2 means high contrast decreases it. $w_5 = 0$ implies no interaction needed regularization pruned it. To test the low predictive value of x_3 : perform a hypothesis test on w_3 or retrain excluding x_3 .

(g) ROC and AUC Evaluation

Scores : positives $\{0.85, 0.60, 0.45\}$,
negatives $\{0.35, 0.25, 0.15, 0.10\}$.

Since every positive score exceeds every negative score, the ROC curve reaches $(FPR = 0, TPR = 1)$, and the area under the curve is 1.0.

Threshold	TPR	FPR	Notes
0.5	2/3	0	Positives with $p \geq 0.5$: $\{0.85, 0.60\}$
0.45	1	0	All positives captured, no FP
0.35	1	1/4	First negative included

Higher λ compresses probabilities toward 0.5, flattening ROC if separation worsen.

(h) Convexity and Hessian

With $S = \text{diag}(p_i(1-p_i))$ and $X \in \mathbb{R}^{N \times d}$

$$\nabla^2 \ell = \frac{1}{N} X^T S X \geq 0$$

Thus ℓ is convex with a unique global minimizer when X has full column rank. Adding Elastic Net keeps the objective convex: the L2 part is smooth and convex, the L1 part is convex but non-smooth, so the total remains convex, and first-order methods with subgradient or proximal steps apply.

(i) Connection to LDA

from Gaussian class condition densities and Bayes' theorem:

$$\ln \frac{P(y=1|x)}{P(y=0|x)} = (\Sigma^{-1}(\mu_1 - \mu_0))^T x + \text{const}$$

Thus both LDA and Logistic regression have linear decision boundaries $w^T x + b = 0$. LDA is generative, while logistic regression is discriminative. When class-conditional features are approximately Gaussian, logistic regression approximates the optimal Bayes boundaries.